

# Kuramoto Model of Brain Entropy Under Psychedelics

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## Introduction

N,N-Dimethyltryptamine (DMT) is a naturally occurring indolealkylamine that has been found to possess potent psychoactive properties. It is known to induce an intense, but short-lived, hallucinogenic experience characterized by vivid visual imagery, altered sense of time and space, and profound changes in mood and perception (Strassman et al., 1994). While DMT has been used for centuries in various indigenous Amazonian shamanic practices, such as in the brew ayahuasca (Rafael et al., 2011). Its therapeutic potential and the neuroscientific underpinnings of its effects have only recently begun to be explored.

Resurgence of research on psychedelic compounds has highlighted the potential of DMT as a novel treatment for various mental health conditions. For instance, ayahuasca, which contains DMT as its main active ingredient, has shown promise in the treatment of depression (Osório et al., 2015) and substance use disorders (Lucas Silva Rodrigues et al., 2021). However, we still lack a clear understanding of the neural mechanisms underlying DMT's effects.

Computational neuroscience allows for modeling and predicting the complex neural dynamics induced by DMT. By combining neuroimaging data with advanced computational techniques, such as dynamic causal modeling (Suresh Muthukumaraswamy et al., 2013) and whole-brain network analysis (Deco et al., 2018). Research has mostly converged on the idea that the psychedelic brain state is caused by the dysregulation of higher order networks, specifically the Default Mode Network (DMN) with decreased-within network integration (Timmermann et al., 2023), being linked to the breakdown of segregated networks, causing an increase in the global functional connectivity in the absence of the top-down regulation the DMN normally enforces on these networks, leading to a state described in as an “anarchic brain” in the REBUS model of brain action under psychedelics (Carhart-Harris & Friston, 2019).

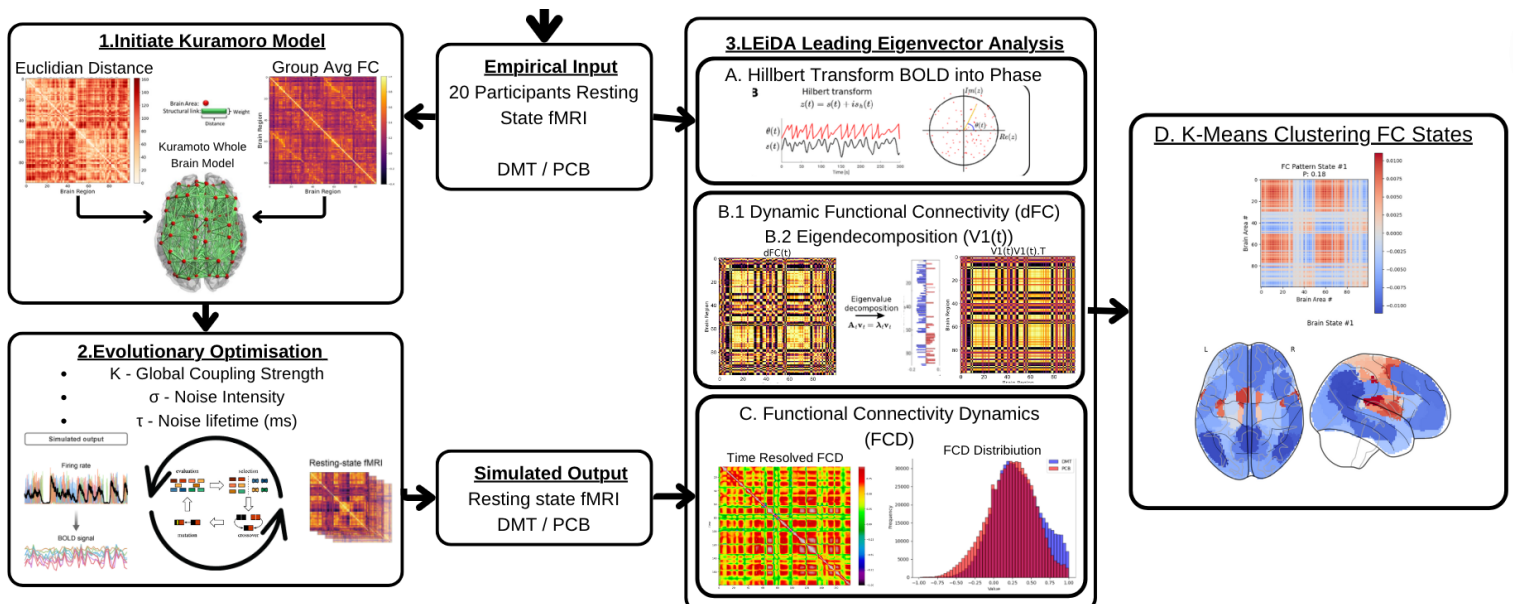
In this paper, we aimed to evaluate how well the Kuramoto oscillator model can simulate the brain states induced by DMT. We applied the Leading Eigenvector Analysis (LEIDA) technique (Cabral et al., 2017) to fMRI recordings of brains under DMT compared to placebo (PCB). This study generated computational models for DMT and PCB, processed the data using LEIDA to evaluate Functional Connectivity Dynamics (FCD), and assessed the Kuramoto model's ability to simulate functional connectivity states.

## Methods.

### Procedure

The study involved 20 healthy volunteers (mean age = 33.5 years, SD = 7.9; 7 females) in a single-blind, placebo-controlled, counterbalanced, within-subjects design. Each participant attended two sessions with a two-week interval, receiving either a placebo (10 mL saline) or 20 mg DMT (in 10 mL saline). Resting-state fMRI data were acquired with a 3T Siemens MR B17 scanner, and preprocessing included despiking, motion correction, spatial smoothing, and band-pass filtering. The Schaefer atlas, dividing the brain into 100 regions, was used for parcellation, focusing on the first 6 minutes post-administration for network dynamics analysis.

**Figure 1**  
*Methods diagram of approach*



**Figure 1.** Figure illustrating the methodological approach. Functional connectivity matrices were extracted from empirical BOLD recordings for all individual in a group. Average group FC input to Kuramoto model (1). Then BOLD output optimised to maximise FC fit to each individuals FC matrix via evolutionary optimisation (2). After which all data from empirical and simulated enters the same LEiDA analysis (3).

**Static Functional Connectivity:**

Static Functional Connectivity (sFC) was calculated as the pairwise Pearson correlation of BOLD activity across all time points ( $T = 180$ ), for each of the regions ( $N = 100$ ), resulting in a single  $N \times N$  matrix representing functional brain connectivity.

**Dynamic Functional Connectivity:**

Dynamic Functional Connectivity (dFC) matrices were computed from the phase coherence of BOLD activity across time. The BOLD signal was first converted into phase using the Hilbert transform, unwrapping the signal amplitude from phase, yielding  $\theta(n, t)$  for each region  $n$  at each timepoint  $t$ . The phase coherence between regions  $n$  and  $p$  at time  $t$  was then calculated as dFC using the following equation:

*Equation.1*

$$dFC(n, p, t) = \cos(\theta(n, t) - \theta(p, t))$$

**Eigen Decomposition of dFC:**

Eigenvalue decomposition is a mathematical technique used to factorise a matrix into a set of orthogonal eigenvectors and their corresponding eigenvalues. When applied to dFC matrices (Eq.2), eigenvalue decompositions at each time point represent the the dominant patterns of functional connectivity. It is a dimensionality reduction technique that becomes useful in dynamic analysis for its ability to capture dominant FC brain states whilst maintaining around 70% of variance. For each time point  $t$ , the  $dFC(t)$  matrix, which is an  $N \times N$  symmetric matrix, is decomposed into a set of  $N$  eigenvectors, denoted as  $V(t)$  and their associated eigenvalues, denoted as  $\lambda(t)$ . The eigenvectors are ordered based on the magnitude of their corresponding eigenvalues, with  $V_1(t)$  being the leading (negative) eigenvector, representing the dominant functional connectivity pattern of the  $dFC(t)$  matrix at time  $t$ .

*Equation.2*

$$dFC(t) \cdot v_i(t) = \lambda_i(t) \cdot v_i(t)$$

**Functional Connectivity Dynamics (FCD):**

The Functional Connectivity Dynamics (FCD) matrix is computed to investigate the temporal evolution of the dominant connectivity patterns captured by the dFC leading eigenvectors. The FCD matrix quantifies the similarity between the leading eigenvectors  $V_1(t)$  at different time points, revealing how the principal connectivity patterns change over time. The FCD matrix is calculated as the cosine similarity between pairs of leading eigenvectors (Eq.3). The cosine similarity measures the cosine of the angle between two vectors and provides a bounded similarity measure between -1 and 1. The FCD matrix is a symmetric  $T \times T$  matrix, where each entry  $FCD(t_x, t_y)$  represents the similarity between the leading eigenvectors at time points  $t_x$  and  $t_y$ .

*Equation.3*

$$FCD(t_x, t_y) = \frac{V_1(t_x) \cdot V_1(t_y)}{\|V_1(t_x)\| \|V_1(t_y)\|}$$

**Kuramoto Modelling:**

The Kuramoto model, developed by Yoshiki Kuramoto (1973), is described by a differential equation (Eq. 4) governing the evolution of phase ( $\theta$ ) for each oscillator in a network. The model accounts for intrinsic frequencies ( $\omega_i$ ), noise ( $\zeta_i$ ) generated by an Ornstein-Uhlenbeck (OU) process (Eq. 5) characterized by noise intensity ( $\sigma$ ) and noise decay ( $\tau$ ), global coupling strength ( $K$ ), adjacency matrix ( $A$ ) indicating inter-node connectivity, and transmission delays ( $\tau$ ) affecting phase synchronization dynamics.

*Equation.4*

$$\frac{d\theta_i}{dt} = \omega_i + \zeta_i + \frac{K}{N} \sum_{j=1}^N A_{ij} \sin(\theta_j(t - \tau_{ij}) - \theta_i(t))$$

*Equation.5*

$$d\zeta(t) = -\frac{\zeta(t)}{\tau} dt + \sigma$$

**Kuramoto Modelling Application:**

To simulate brain activity under distinct physiological states, separate Kuramoto models were employed for DMT and PCB conditions. These models were initialized using empirical data from each group and optimized using the Neurolib framework (Caglar Cakan et al., 2021). Initialization parameters included Euclidean distance-based delays ( $\tau$ ) and group-averaged sFC served as input for coupling strength ( $A$ ). A uniform transmission speed ( $V = 20$ ) was used, with initial phases randomly sampled from a normal distribution and a fixed intrinsic node frequency of 0.08.

**Simulation and Optimization:**

The models simulated theta rhythms over 6 minutes, producing BOLD signals through the Balloon-Wagner model, which converts theta rhythms into BOLD signals. These were down-sampled to 0.5 Hz and normalized (Z-scored) to yield 100 x 180 BOLD time series, aligning with empirical data. To optimize model fidelity to empirical sFC, an evolutionary algorithm (NSGA2) iterated over 20 generations with a population size of 20. Parameters optimized included coupling strength ( $K$ , 1 to 10), noise intensity ( $\sigma$ , 0.01 to 0.5), and decay time ( $\tau$ , 1 to 100 ms). NSGA2 optimized parameters to create sFCs that maximally fit individual empirical sFC profiles within each group, enhancing model representativeness.

## Results

### Optimization:

The DMT model exhibited a higher global coupling strength ( $K$ ) and a significantly longer noise lifetime ( $\tau$ ) compared to the PCB model. Despite the PCB model having greater noise intensity ( $\sigma$ ), these findings suggest that the DMT neural state has higher global connectivity, prolonged noise propagation, and increased entropy compared to the PCB neural state.

**Table 1**

Evolutionary Optimal Parameters

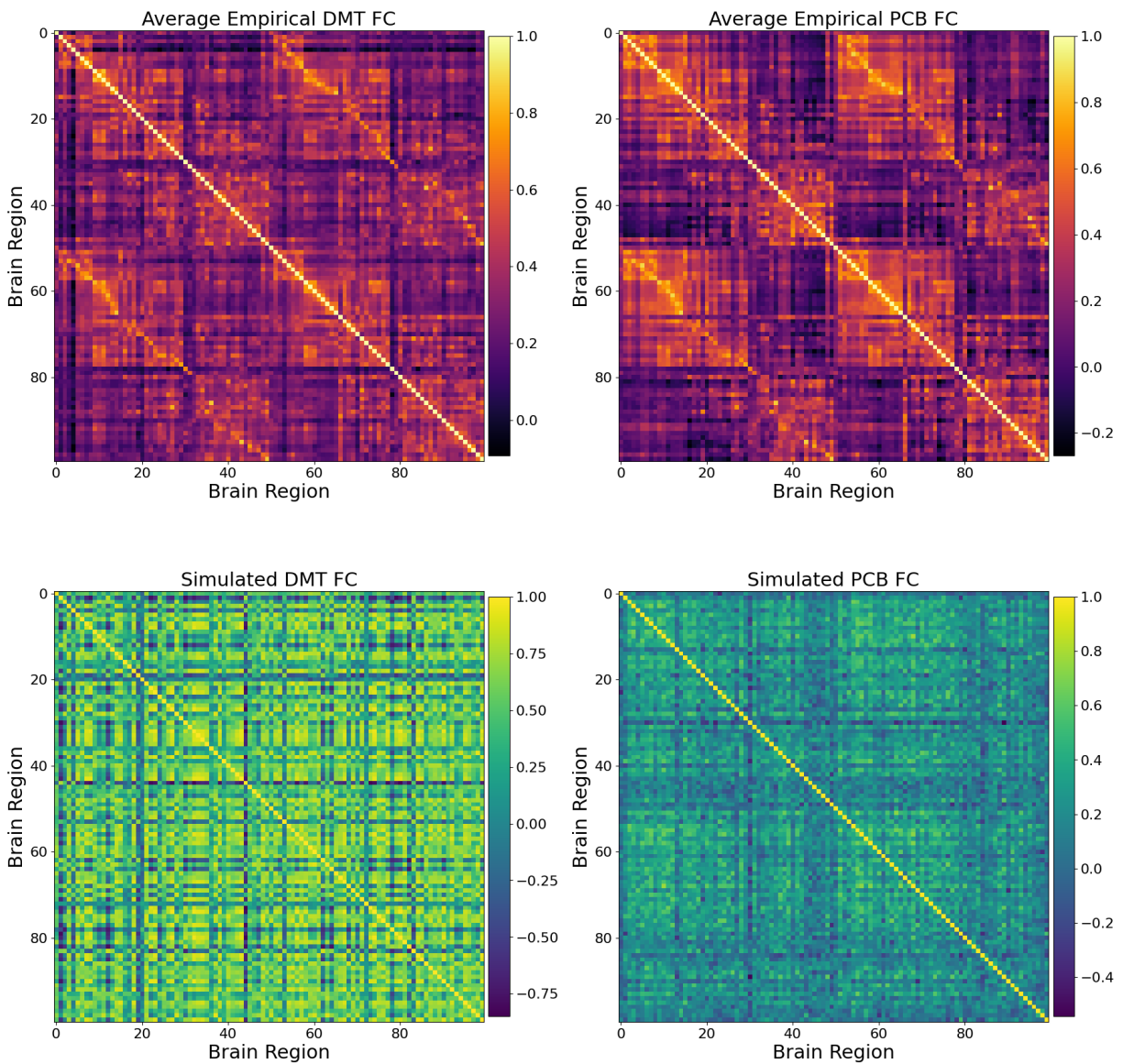
| Model   | $K$  | Noise Lifetime ( $\tau$ /ms) | Noise Intensity ( $\sigma$ ) |
|---------|------|------------------------------|------------------------------|
| DMT     | 8.87 | 69.62                        | 0.17                         |
| Control | 6.03 | 9.03                         | 0.34                         |

*Note:* Where  $K$  is global coupling,  $\tau$  lifetime of noise and  $\sigma$  is the level of noise

### Static Functional Connectivity:

Empirical and simulated sFC are shown in figure 2. In the DMT condition, the mean correlation between the simulated sFC and individual empirical sFCs was 0.14 (SD = 0.05), with a correlation of 0.21 to the group-averaged empirical sFC. Conversely, the PCB condition exhibited a slightly better fit, with a mean correlation of 0.19 (SD = 0.06) between the simulated sFC and individual empirical sFCs, and a correlation of 0.29 to the group-averaged empirical sFC.

**Figure 2.** *Static Functional Connectivity Matrices*



*Figure 2.* Pearsons's correlation of node-to-node FC across the duration of BOLD for DMT (left), Placebo (right), with the simulated being in green (bottom) and group averaged empirical being in red(top).

### **FCD Comparison:**

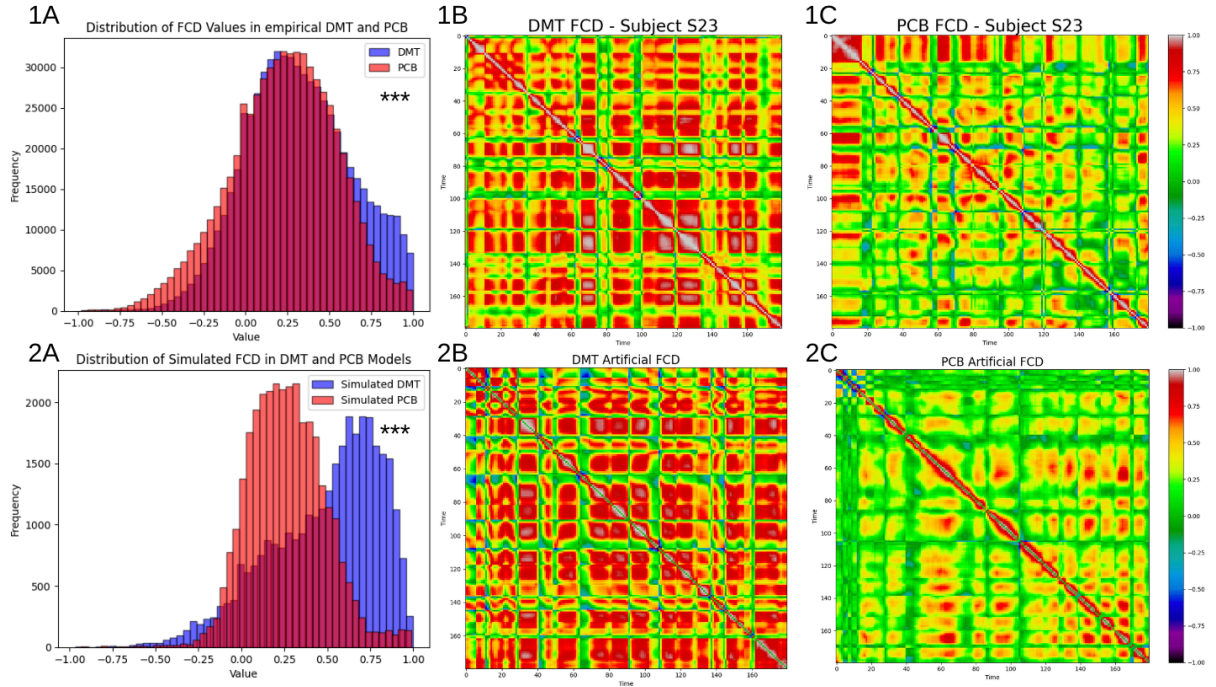
The FCD analysis (Fig.3) reveals the dynamic state transitions and entropy aspects, with an increased number of state shifts indicating faster switches between brain states. The DMT condition, as visualized for participant S23 (1B), displays a high similarity in transient FC states across time. This indicates a heightened global synchronization, contrasting with the PCB (1C) condition where states transitions are more gradual with less reoccurrence reflecting stable connectivity.

Empirical data show a shift in FCD value distributions (1A) under DMT towards higher temporal similarity, compared to PCB, suggesting an overall increase in whole-brain connectivity dynamics. In contrast, the placebo condition exhibits more fragmented connectivity patterns.

Simulated models generally mimic empirical observations but vary in accuracy; the Kumamoto model, while visually similar to the DMT data, overestimates temporal coupling, showing a pronounced Poisson shift towards higher values. The PCB model aligns more closely with empirical data, accurately representing the more stable connectivity observed however exhibiting less anti-correlations.

In summary, FCD analysis underscores that DMT significantly enhances global connectivity synchronization compared to PCB. While the models capture key aspects of the DMT-induced connectivity dynamics, they require refinement to more precisely mirror the empirical data's observed alterations in whole-brain connectivity.

**Figure 3.** *Functional Connectivity Dynamics (FCD) Matrices*

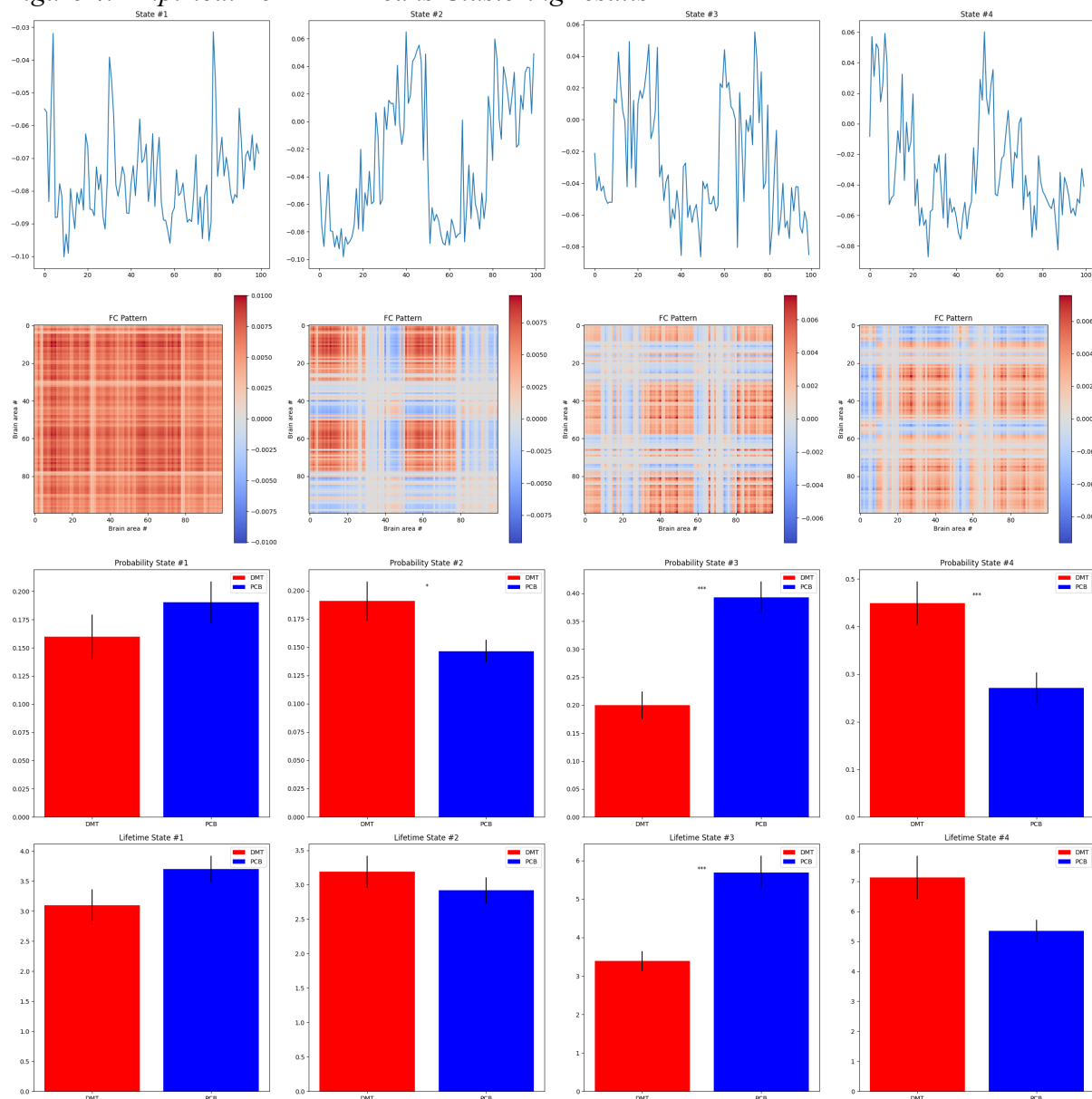


*Figure 3.* Time resolved Functional Connectivity Dynamics (FCD) plots of one example participant empirical recordings in both DMT (1B) and PCB (1C) states. Simulated FCD plots (2B:2C). Left distribution frequency plots of lower triangle flattened FCD matrices for all empirical participants (1A) and the two models (2A) with the Kurtosis-Smirnov test comparing distribution. *Note: \*\*\* denotes  $p > .05$*

## Functional Connectivity States:

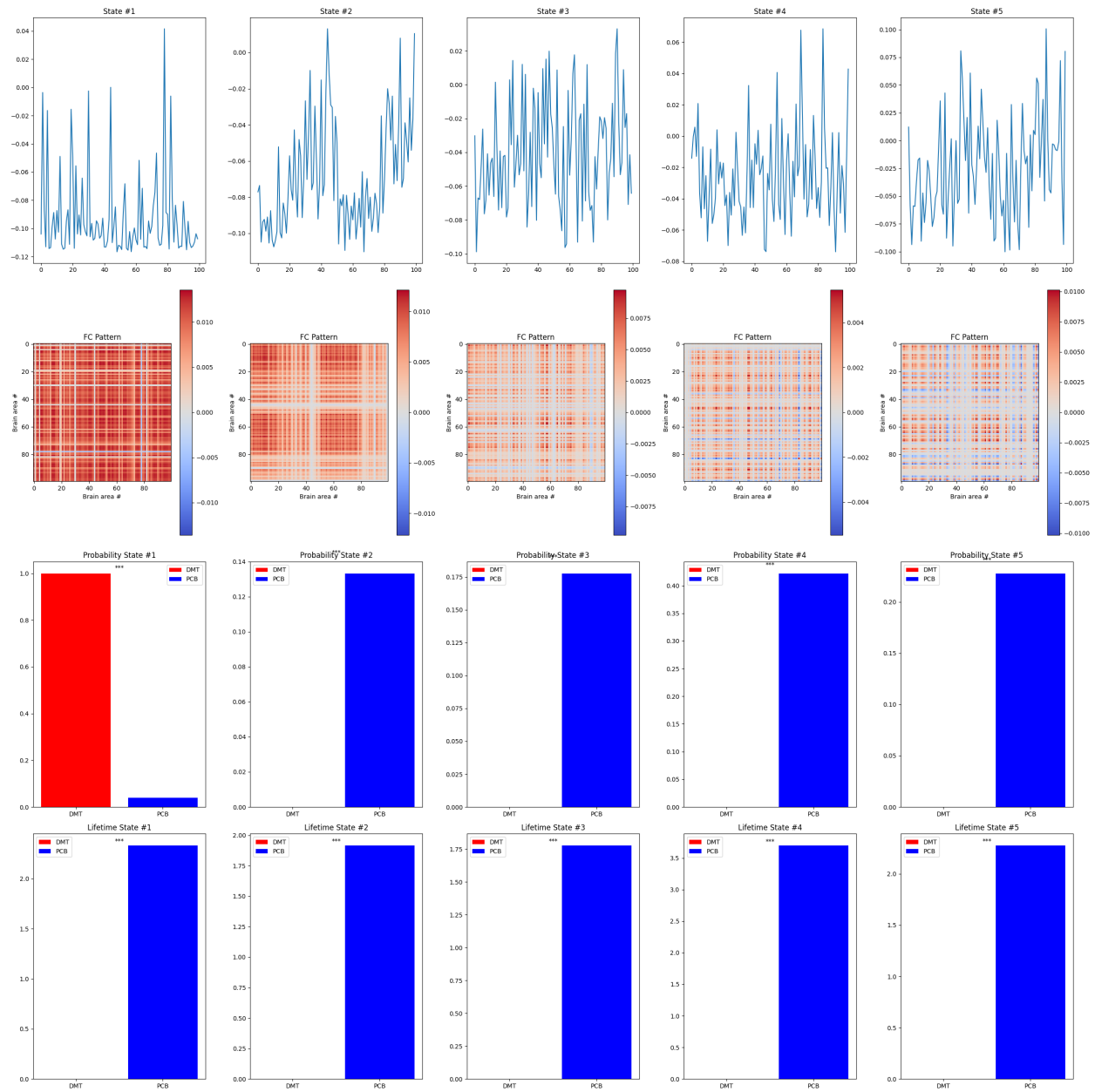
The K-Means Clustering of eigenvectors can identify reoccurring FC states, describe their probability and duration. For simplicity we chose the first 4 clusters.

*Figure 4. Empirical LeIDA K-Means Clustering results*



*Note: Probability is the percentage (/100) change of that state occurring out of all participants. Lifetime is the average duration of that state (in seconds). DAN [(16, 23), (67, 73)], DMN [(38, 50), (90, 100)], FPN [(34, 37), (81, 89)], LIN [(31, 33), (79, 80)], SAN [(24, 30), (74, 78)], SMN [(10, 15), (59, 66)], VIN [(1, 9), (51, 58)]*

Figure 5. Simulated LeIDA K-Means Clustering results



## Discussion

Our analysis of empirical data revealed four distinct functional connectivity (FC) states, each with unique characteristics. The second FC state, which was more prevalent in DMT brains, showed an increased contribution of lower-order networks, including visual and somatosensory networks, and was related to the Trans Model-Association Pole (TOP) (Timmermann, 2020). In contrast, the third state, which was more likely to occur in PCB and lasted longer, was characterized by the synchronized contribution of higher-order networks, including the default mode network (DMN) and frontoparietal network (FPN), with some visual activation, typical of the resting state neural state. The lower probability of this state in DMT brains provides evidence for DMN dissociation under DMT (Carhart-Harris, 2018).

The final FC state showed a global decrease in between-network segregation, with simultaneous activation of multiple networks, including the SMN, SAN, LIN, FPN, and DMN. This is consistent with the literature, which suggests that the psychedelic state is characterized by a decrease in between-network segregation (Carhart-Harris & Friston, 2019).

Our clustering analysis on simulated data did not yield meaningful results, likely due to the models being initiated without biological realism. However, the models were able to replicate the functional connectivity dynamics (FCD) of empirical data, particularly the PCB model. Optimization parameters provided computational evidence to support the entropic brain theory, which suggests that the psychedelic brain state is caused by increased entropy in the DMN, leading to downstream disinhibition of lower-order networks (Carhart-Harris et al., 2014).

Future studies can apply the Kuramoto model to provide further evidence for the entropic brain theory. By establishing a model with the typical resting-state FC pattern and selectively adding noise to the DMN nodes via an Ornstein-Uhlenbeck (OU) process, we can determine if DMN entropy is truly causal to the psychedelic state observed, similar to what Alves et al., (2022) did using Machine learning to categorise DMT induced changes in EEG signal using a network approach.

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K-Means Eigen: <https://www.frontiersin.org/articles/10.3389/fnhum.2022.980280/full>

Nice Graphic pcilocibin!/: [https://www.researchgate.net/figure/Methodological-pipeline-for-leading-eigenvector-dynamics-analysis-LEiDA-and-diametrical\\_fig1\\_357153896](https://www.researchgate.net/figure/Methodological-pipeline-for-leading-eigenvector-dynamics-analysis-LEiDA-and-diametrical_fig1_357153896)

Improved Lieda:

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